Enhancing Endoscopic Diagnosis of GERD and Polyps: A Comparative Study of Deep Learning Models for Multi-Class Classification

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Abstract—Accurate and timely diagnosis of gastroesophageal disease and polyps from endoscopic images remains a demanding but crucial function in clinical practice due to visual analogy between pathological and non-pathological tissues. In this research, comparative evaluation of three deep learning models, ResNet-50, InceptionV3, and Ensemble Transfer Learning (Ensemble TL) framework, for multi-class classification of GERD, GERD Normal, Polyp, and Polyp Normal using high-resolution endoscopic images is presented. The models were evaluated based on performance metrics: accuracy, precision, recall, F1 measure, and classification accuracy. ResNet-50 showed strong baseline performance with an accuracy of 73% and balanced performance on Polyp detection (F1: 0.78). Its GERD recall (0.65) and GERD Normal precision (0.64) reflected challenges in separating closely similar visual features. InceptionV3 achieved improved performance, particularly in GERD (F1: 0.78) and Polyp (F1: 0.83) detection, with a overall accuracy of 77%. It did degrade its performance in GERD Normal sensitivity, though. The Ensemble TL model outperformed all the individual models, with a highest accuracy of 87% and achieving improved F1 scores for all disease classes: GERD (0.85), GERD Normal (0.82), Polyp (0.94), and Polyp Normal (0.87). These enhancements are credited to the ability of the model to integrate high-level features from ResNet and Inception architectures, enhancing global and local feature learning. The paper underlines ensemble learning's role in complex medical imaging applications, giving consistent and transferable performance across diverse forms of diseases.

Keywords—Gastroesophageal Disease, Medical Imaging, Transfer Learning, Ensemble Model

I. Introduction

Gastrointestinal (GI) diseases are among the leading causes of morbidity and mortality globally, with significant clinical and economic burdens [1]. Two conditions of particular concern are Gastroesophageal Reflux Disease (GERD) and colorectal polyps, which, if left undiagnosed or untreated, can lead to serious complications such as Barrett's esophagus, esophageal adenocarcinoma, and colorectal cancer. GERD is a chronic digestive disorder resulting from the backflow of stomach acid into the esophagus, manifesting through symptoms like heartburn, chest pain, regurgitation, and inflammation. It affects an estimated 20% of adults in Western countries, and its prevalence is rising globally due to dietary changes, obesity, and sedentary lifestyles [2]. Meanwhile, colorectal polyps—small clumps of cells forming on the lining of the colon or rectum—are often asymptomatic but can become precancerous if not detected early. Alarmingly, colorectal cancer ranks third in global cancer incidence, accounting for over 1.9 million new cases and nearly 935,000 deaths in 2022 alone [3].

With the rise of deep learning in medical image analysis, Convolutional Neural Networks (CNNs) have demonstrated robust performance in automated diagnostic systems. Models such as ResNet50 and InceptionV3 are widely recognized for their feature extraction capabilities and are frequently applied in radiological and pathological domains [4] – [6]. However, comparative studies on their efficacy for gastrointestinal image classification particularly for GERD and polyp detection remain limited.

In this study, we used a curated dataset of 4,006 high-resolution endoscopic images obtained from Zainul Haque Sikder Women's Medical College & Hospital. We investigate and compare the performance of ResNet50, InceptionV3, and their ensemble architecture for classifying images into four categories: GERD, GERD Normal, Polyp, and Polyp Normal. Our aim is to provide a foundational benchmark for CNN-based diagnostic systems in gastroenterology and assess the clinical relevance of feature-level fusion in ensemble models.

II. Related Works

Deep learning is now a crucial method in early colorectal neoplasms' detection and diagnosis, particularly in colonoscopy polyp detection. The method employs complex algorithms to achieve maximum detection rates for polyps and the quality of diagnostics and hence significantly advances the prevention of

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colorectal cancer [7] - [9]. Current literature emphasizes shows of real-time deep learning algorithms in the detection of polyps during colonoscopies through stunning performances. For instance, Quan et al. [7] reported that their AI-driven polyp detection system was able to detect endoscopists' identified polyps in clinical trials. Wesp et al. [10] demonstrated that deep learning can discriminate between premalignant adenomas and benign hyperplastic polyps, once again underlining the importance of correct classification in avoiding colorectal cancer. Wang et al. [9] also mentioned the improvement in the detection rate of polyps through the implementation of real-time automatic detection systems in clinical practice. Deep learning techniques are also used in various algorithmic approaches and models, which have contributed significantly to enhanced detection performance. For example, several research papers have utilized architecture such as YOLO (You Only Look Once) for real-time detection [11] [12]. Faster R-CNN development has also been promising for the detection and segmentation of polyps in an efficient manner [13]. Research has supplemented the method with solutions to classification tasks, showing that there is a potential accuracy level of up to 80% with acceptable precision and recall scores for adenomatous polyp detection [14]. Integration of deep learning architectures, such as multi-scale hybrid networks, has led to improved detection in various contexts, such as wireless capsule endoscopy. Solutions such as transfer learning have subsequently emerged to counter this difficulty, fine-tuning pre-existing models for polyp detection with limited training resources. Evolution of deep learning in the detection of polyp is revolutionizing approach to enhance early detection and management of colorectal cancer, ultimately improving the outcome of patient by early intervention of medical care and increased rates of detection during routine screening. Even there is still gap to identify disease like Gerd, polyp in real-time scenario

III. Methodology

The proposed system for gastrointestinal disease classification employs a dual-stream ensemble architecture that integrates ResNet50 and InceptionV3 for robust feature learning and decision fusion. Initially, raw endoscopic images are collected and preprocessed through resizing, cropping, normalization, and data augmentation to enhance generalization and model compatibility. These prepared images are independently passed through both pre-trained CNN models, where spatial and hierarchical features are extracted. A late-fusion strategy combines the output predictions from each model, enabling improved classification accuracy across multiple classes, including normal and abnormal gastric polyp and Gerd conditions. The system performance is visualized using metrics such as accuracy, F1-score, and confusion matrices. Figure 1 presents a schematic view of the proposed ensemble-based architecture.

A. Dataset Description

Our dataset consists of four classes: normal and disease categories for Z-line and polyps. The dataset utilized in this study originates from Zainul Haque Sikder Women's Medical College & Hospital (Pvt.) Ltd., with 4,006 original endoscopic images labeled into four categories: GERD, GERD Normal, Polyp, and Polyp Normal [15]. Figure 2 shows our dataset sample images.

B. Data Preprocessing

Preprocessing is an essential step to improve the performance and convergence rate of deep learning models. In this research study, we employed a comprehensive approach to preprocess the gastrointestinal endoscopy dataset to train our models. Images were resized and cropped to match model input, augmented to reduce overfitting, normalized using ImageNet stats, split into train/validation/test sets, and balanced with SMOTE.

C. Proposed Ensemble Model

In this study, we propose an ensemble deep learning approach that combines the potential of ResNet50 and InceptionV3, two of the top convolutional neural networks (CNNs) designed for image classification problems. The two base models were first pre-trained with ImageNet pre-trained weights and then fine-tuned on our carefully prepared gastrointestinal endoscopy dataset. To capitalize on their complementary feature extraction capabilities, we adopted a soft-voting ensemble strategy to aggregate the prediction probabilities from each model. The final ensemble prediction P ensemble computer as:

$$P_{\text{ensemble}} = \alpha \times P\,ResNet50 + (1 - \alpha) \times PInceptionV3 \quad (1)$$

Here, α is an empirically obtained weight factor, which here is assigned the value of 0.5 to equally prefer both models.

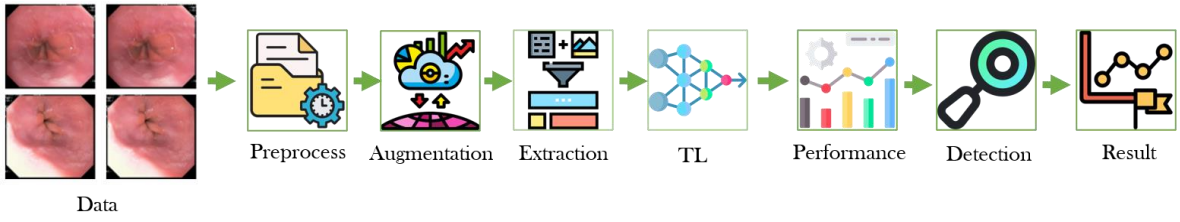


Fig. 1. Our step-by-step procedure diagram.

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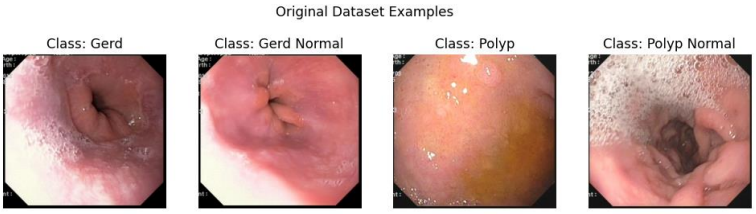


Fig. 2. Our Dataset Example.

D. Performance Evaluation Matrix

We evaluated performance using various metrics to identify the most effective classifier for detecting gastric diseases. Performance indicators, expressed as percentages (%), were calculated using Eqs. (2–5). Additionally, we generated a confusion matrix for each model to assess their performance comprehensively.

$$Accuracy = \frac{True\ Positive + True\ Negative}{All\ Instance} \quad (2)$$

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad (3)$$

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad (4)$$

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (5)$$

IV. Result Analysis and Discussion

The research presents a comparative analysis of deep learning models, ResNet-50, InceptionV3, and Ensemble Transfer Learning for the diagnosis of gastroesophageal and polyp diseases from endoscopic image data. All models were comprehensively tested using precision, recall, F1 score, and accuracy to evaluate their performance individually and on a per-class basis within 20 epoch.

The ResNet-50 model was moderately performing with an overall accuracy of 73% shows in Table I. It had good detection capability for the "Polyp" class with balanced metrics on precision, recall, and F1 score (0.78). GERD detection was quite weaker, though, with a recall of just 0.65 with good precision of 0.80. GERD Normal class followed a reverse pattern with high recall (0.78) but poor precision (0.64), indicating class confusion most likely due to feature overlap. Similarly, Polyp Normal performed moderately and balanced with an F1 score of 0.71. ResNet50's varying accuracies over the different classes shown in figure 3. It is most accurate in correctly classifying 'Polyp' and 'Gerd'. There is, nevertheless, some confusion between 'Gerd' and 'Gerd Normal', and a considerable number of 'Polyp Normal' cases are misclassified as 'Polyp'. This shows that the model is struggling to distinguish between these classes. The model shows good early learning and generalization. The plateauing and increase towards the end in validation loss that it is experiencing some amount of overfitting can be seen in figure 4.

TABLE I. Performance of ResNet50 Model

Model	Class	Precision	Recall	F1	Accuracy
ResNet-50	Gerd	0.80	0.65	0.72	73
	Gerd Normal	0.64	0.78	0.70	
	Polyp	0.78	0.78	0.78	
	Polyp Normal	0.70	0.72	0.71	

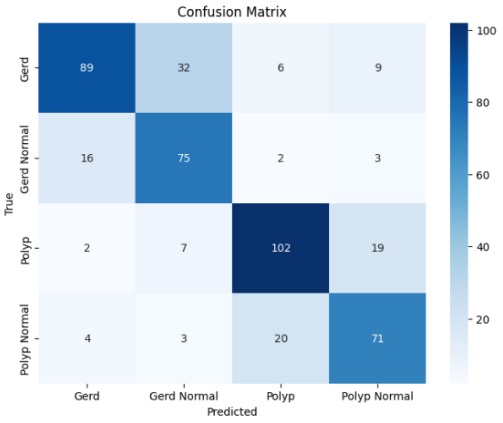


Fig. 3. Confusion Matrix of ResNet50.



Fig. 4. Training and Validation Loss/ Accuracy of ResNet50.

TABLE II. Performance of InceptionV3 Model

Model	Class	Precision	Recall	F1	Accuracy
Inception-V3	Gerd	0.77	0.80	0.78	77%
	Gerd Normal	0.83	0.61	0.71	
	Polyp	0.84	0.82	0.83	
	Polyp Normal	0.68	0.84	0.75	

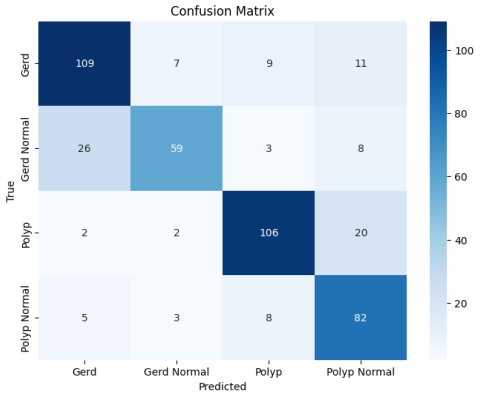


Fig. 5. Confusion Matrix of InceptionV3.

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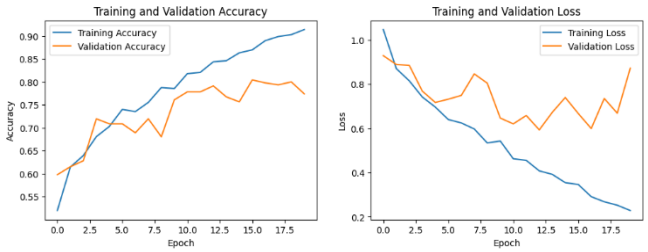


Fig. 6. Training and Validation Loss/ Accuracy of InceptionV3.

From Table II, we can see InceptionV3 performance. The InceptionV3 model outperformed ResNet-50 by an accuracy of 77%. It improved GERD recall to 0.80 with a still high F1 score of 0.78, demonstrating improved sensitivity to GERD markers. Furthermore, polyp classification improved to an F1 of 0.83. However, Normal recall for GERD decreased to 0.61 despite increased precision (0.83) and represented a trade-off between specificity and sensitivity. Normal performance for polyp was improved (F1: 0.75), presumably due to InceptionV3's deeper structure and multi-scale feature extraction. In figure 5 shows that InceptionV3 confusion matrix has a mixed bag of results compared to the ResNet50. The model has improved in its ability to correctly classify 'Gerd' and 'Polyp Normal', whereas its ability to correctly classify 'Gerd Normal' has been lost, confusing it more with 'Gerd'. Performance for 'Polyp' remains good as like ResNet50. The model is fitting the training data very well, but it is showing clear signs of overfitting shown in figure 6. The validation performance, after a certain point, is getting volatile and even worsening as the training performance continues to improve.

TABLE III. PERFORMANCE OF ENSEMBLE MODEL

Model	Class	Precision	Recall	F1	Accuracy
Ensemble TL	Gerd	0.95	0.77	0.85	87%
	Gerd Normal	0.72	0.95	0.82	
	Polyp	0.98	0.89	0.94	
	Polyp Normal	0.86	0.92	0.87	

Ensemble Transfer Learning performance shows in Table III and it that has the advantage of both ResNet50 and InceptionV3 greatly outperformed the individual models with a combined accuracy of 87%. GERD classification much improved to a 0.95 precision and an F1 of 0.85 showing not only better sensitivity but also fewer false positives. GERD Normal classification was also improved (Recall: 0.95, F1: 0.82), overcoming the previous failures seen in the individual models. Polyp classification registered the highest F1 score at 0.94, followed by Polyp Normal performing well (F1: 0.87), demonstrating the capacity of the ensemble model to generalize to varying and minor tissue presentations. Ensemble model confusion matrix shown in figure 7 and it demonstrates overall good performance across all categories. The model is particularly good at classifying 'Gerd Normal' and 'Polyp' with very few misclassifications. Although the 'Gerd' class has more misclassifications as 'Gerd Normal', and 'Polyp Normal' is misclassified on several other classes, the overall results reflect a well-performing model on this data. The model shows good early learning and generalization while the model is overfitting on the training data. It achieves very high accuracy and low loss on the training set, its performance on the validation set has plateaued at a much lower-level shows in figure 8.

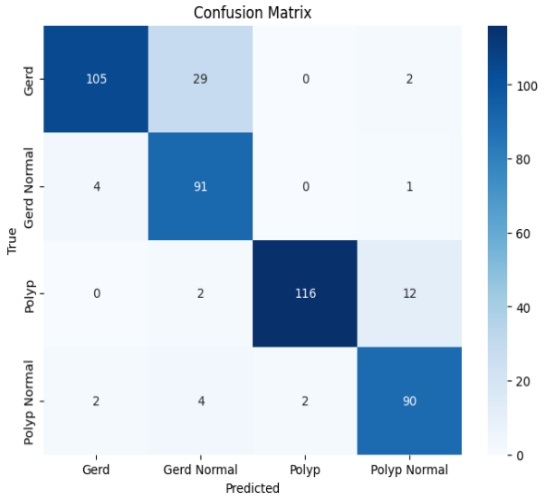


Fig. 7. Confusion Matrix of Ensemble Model.

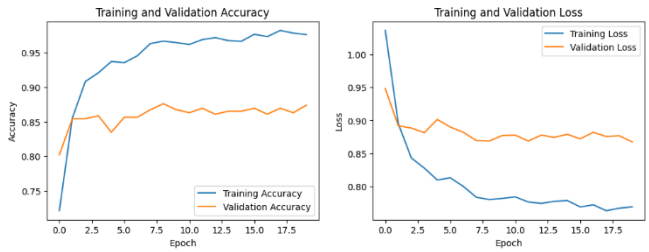


Fig. 8. Training and Validation Loss/ Accuracy of Ensemble Model.

From Table IV, out of the three models tested, ResNet-50, Inception-V3, and Ensemble TL, the Ensemble TL model did much better than standalone architectures, particularly class-wise balance, generalization, and overall reliability. While ResNet-50 and Inception-V3 were deficient in class-specific flaws such as low recall on Gerd and Gerd Normal, the Ensemble model exhibited top F1 values for all the classes, particularly noting 0.94 on Polyp and 0.85 on Gerd, which are crucial to enable early detection and intervention. Although a minor problem of overfitting during training was encountered, the ensemble approach properly leveraged the advantages of both ResNet and Inception-V3 without their respective disadvantages.

TABLE IV. OVERALL MODEL REMARKS

Model	**Best Class**	**Weakness**	**Overall Remarks**
ResNet-50	Polyp	Low recall for Gerd (0.65)	Struggles with Gerd Identification
Inception-V3	Polyp	Lower recall for Gerd Normal (0.61)	Better generalization than ResNet-50
Ensemble TL	Polyp	Slightly lower precision for Gerd Normal	Superior performance overall

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V. Conclusion

This study analyzed the efficacy of deep convolutional neural network architectures such as ResNet-50, InceptionV3, and their combination as an Ensemble Transfer Learning model on endoscopic disease classification. All models presented distinct strengths where ResNet-50 delivered persistent but poor performance and InceptionV3 achieved considerable improvements on GERD and Polyp classification. However, both standalone models struggled with inter-class discrimination, particularly for GERD Normal, where precision or recall was traded due to delicate visual overlaps. The Ensemble TL model overcame these drawbacks through a strategic blend of ResNet's residual learning capability and InceptionV3's multi-scale feature management. The hybrid network worked best in optimizing precision and recall for all four classes at the same time, with an impressive overall. The robustness of the ensemble model, its high F1 scores, and capacity to generalize across all disease types recommend its clinical use in resource-limited or high-throughput screening settings but overfitting in training condition which we will improve with L1 and L2 regularization.

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